

Program to reverse a string using different methods (menu-driven)

```
def reverse_slicing(s):  
    return s[::-1]
```

```
def reverse_loop(s):  
    reversed_str = ""  
    for char in s:  
        reversed_str = char + reversed_str  
    return reversed_str
```

```
def reverse_builtin(s):  
    return "".join(reversed(s))
```

```
while True:  
    print("\n===== STRING REVERSAL MENU =====")  
    print("1. Reverse using Slicing")  
    print("2. Reverse using Loop")  
    print("3. Reverse using reversed() function")  
    print("4. Exit")  
    print("=====")
```

```
choice = int(input("Enter your choice (1-4): "))
```

```
if choice in (1, 2, 3):  
    string = input("Enter a string: ")
```

```
    if choice == 1:  
        print("Reversed string:", reverse_slicing(string))  
    elif choice == 2:  
        print("Reversed string:", reverse_loop(string))
```

```
elif choice == 3:
    print("Reversed string:", reverse_builtin(string))

elif choice == 4:
    print("Exiting program... Goodbye!")
    break
else:
    print("Invalid choice! Please enter a number between 1 and 4.")
```

Program to demonstrate the usage of Bitwise Operators using a menu-driven program

```
def bitwise_and(a, b):
    print(f"{a} & {b} = {a & b}")

def bitwise_or(a, b):
    print(f"{a} | {b} = {a | b}")

def bitwise_xor(a, b):
    print(f"{a} ^ {b} = {a ^ b}")

def bitwise_not(a):
    print(f"~{a} = {~a}")

def bitwise_left_shift(a, n):
    print(f"{a} << {n} = {a << n}")

def bitwise_right_shift(a, n):
    print(f"{a} >> {n} = {a >> n}")
```

Main Program

while True:

```
    print("\n===== BITWISE OPERATORS MENU =====")
```

```
    print("1. Bitwise AND (&)")
```

```
    print("2. Bitwise OR (|)")
```

```
    print("3. Bitwise XOR (^)")
```

```
    print("4. Bitwise NOT (~)")
```

```
    print("5. Left Shift (<<)")
```

```
    print("6. Right Shift (>>)")
```

```
    print("7. Exit")
```

```
    print("=====")
```

```
    choice = int(input("Enter your choice (1-7): "))
```

```
    if choice in (1, 2, 3):
```

```
        a = int(input("Enter first number: "))
```

```
        b = int(input("Enter second number: "))
```

```
        if choice == 1:
```

```
            bitwise_and(a, b)
```

```
        elif choice == 2:
```

```
            bitwise_or(a, b)
```

```
        elif choice == 3:
```

```
            bitwise_xor(a, b)
```

```
    elif choice == 4:
```

```
        a = int(input("Enter a number: "))
```

```
        bitwise_not(a)
```

```
    elif choice == 5:
```

```

a = int(input("Enter a number: "))
n = int(input("Enter number of bits to shift left: "))
bitwise_left_shift(a, n)

elif choice == 6:
    a = int(input("Enter a number: "))
    n = int(input("Enter number of bits to shift right: "))
    bitwise_right_shift(a, n)

elif choice == 7:
    print("Exiting program... Goodbye!")
    break

else:
    print("Invalid choice! Please enter a number between 1 and 7.")

```

Program to count vowels in a given string using different methods

```

def method1_dict(string):
    """Count each vowel using a dictionary"""
    string = string.lower()
    vowels = {'a': 0, 'e': 0, 'i': 0, 'o': 0, 'u': 0}
    for ch in string:
        if ch in vowels:
            vowels[ch] += 1
    print("\nVowel counts:")
    for v, c in vowels.items():
        print(v, ":", c)

def method2_count_function(string):
    """Count each vowel using count() method"""
    string = string.lower()

```

```
vowels = "aeiou"
print("\nVowel counts:")
for v in vowels:
    print(v, ":", string.count(v))
```

```
def method3_total_count(string):
    """Count total number of vowels only"""
    string = string.lower()
    vowels = "aeiou"
    count = 0
    for ch in string:
        if ch in vowels:
            count += 1
    print("\nTotal number of vowels:", count)
```

Main Program

while True:

```
    print("\n===== VOWEL COUNT MENU =====")
    print("1. Count vowels using Dictionary method")
    print("2. Count vowels using count() function")
    print("3. Count total number of vowels (loop method)")
    print("4. Exit")
    print("=====")
```

```
    choice = int(input("Enter your choice (1-4): "))
```

```
    if choice in (1, 2, 3):
```

```
        string = input("Enter a string: ")
```

```
        if choice == 1:
```

```
            method1_dict(string)
```

```
elif choice == 2:
    method2_count_function(string)
elif choice == 3:
    method3_total_count(string)

elif choice == 4:
    print("Exiting program... Goodbye!")
    break

else:
    print("Invalid choice! Please enter a number between 1 and 4.")
```

Program to create, concatenate, print and access substring from a string

```
def create_string():
    s = input("Enter a new string: ")
    print("String created successfully!")
    return s

def concatenate_strings(s):
    add_str = input("Enter another string to concatenate: ")
    s = s + add_str
    print("After concatenation:", s)
    return s

def print_string(s):
    print("Current string:", s)

def substring_access(s):
    start = int(input("Enter the starting index: "))
```

```
end = int(input("Enter the ending index: "))
if 0 <= start < len(s) and 0 < end <= len(s) and start < end:
    print("Substring is:", s[start:end])
else:
    print("Invalid index range!")
```

```
# Main Program
```

```
string = ""
```

```
while True:
```

```
    print("\n===== STRING OPERATIONS MENU =====")
    print("1. Create a String")
    print("2. Concatenate another String")
    print("3. Print the String")
    print("4. Access a Substring")
    print("5. Exit")
    print("=====")
```

```
choice = int(input("Enter your choice (1-5): "))
```

```
if choice == 1:
```

```
    string = create_string()
```

```
elif choice == 2:
```

```
    if string == "":
```

```
        print("Please create a string first!")
```

```
    else:
```

```
        string = concatenate_strings(string)
```

```
elif choice == 3:
```

```
    if string == "":
```

```
        print("No string to display! Please create one first.")
```

```

else:
    print_string(string)

elif choice == 4:
    if string == "":
        print("Please create a string first!")
    else:
        substring_access(string)

elif choice == 5:
    print("Exiting program... Goodbye!")
    break

else:
    print("Invalid choice! Please enter a number between 1 and 5.")

```

Program to perform different pattern matching operations using Regular Expressions

```

import re

def search_pattern(string, pattern):
    """Check if pattern exists anywhere in the string"""
    if re.search(pattern, string):
        print("✔ Pattern found in the string!")
    else:
        print("✗ Pattern not found in the string.")

def find_all_matches(string, pattern):
    """Find and display all occurrences of the pattern"""
    matches = re.findall(pattern, string)
    if matches:

```

```
    print("✔ Pattern found!")
    print("All occurrences:", matches)
else:
    print("✗ No matches found.")
```

```
def count_occurrences(string, pattern):
    """Count how many times the pattern appears"""
    matches = re.findall(pattern, string)
    print(f"Total occurrences of pattern: {len(matches)}")
```

```
# Main program
```

```
while True:
```

```
    print("\n===== REGULAR EXPRESSION MENU =====")
    print("1. Search if pattern exists in string")
    print("2. Find all occurrences of pattern")
    print("3. Count number of occurrences")
    print("4. Exit")
    print("=====")
```

```
choice = int(input("Enter your choice (1-4): "))
```

```
if choice in (1, 2, 3):
```

```
    string = input("Enter the main string: ")
    pattern = input("Enter the pattern to search: ")
```

```
    if choice == 1:
```

```
        search_pattern(string, pattern)
```

```
    elif choice == 2:
```

```
        find_all_matches(string, pattern)
```

```
    elif choice == 3:
```

```
        count_occurrences(string, pattern)
```

```
elif choice == 4:  
    print("Exiting program... Goodbye!")  
    break
```

```
else:  
    print("Invalid choice! Please enter a number between 1 and 4.")
```